

# Higher-Order Residue Deficits in Legendre Intervals

Ruqing Chen

GUT Geoservice Inc., Montréal, Canada

ruqing@hotmail.com

Repository: <https://github.com/Ruqing1963/higher-order-residue-deficits>

February 2026

## Abstract

We generalize the Spin Asymmetry Theorem of [1] from quadratic residues to arbitrary  $k$ -th power residues. Let  $n \geq 2$  with  $p = 2n - 1$  prime, and let  $k \geq 2$  with  $k \mid (p - 1)$ . The interior of the Legendre interval  $\mathcal{I}_n = [(n-1)^2, n^2]$  exhibits a *universal phase deficit*: the  $k$ -th power residue class containing the missing residue  $r \equiv 4^{-1} \pmod{p}$  has exactly one fewer representative than each of the other  $k - 1$  classes. We identify the deficient phase explicitly: for any character  $\chi$  of order  $k$ ,  $\sum_{x \in \mathcal{I}_n^\circ} \chi(x) = -\chi(4^{-1})$ . For cubic residues ( $k = 3$ ), the deficient phase is determined by the cubic character of 2 modulo  $p$ . For quartic residues ( $k = 4$ ), the deficient phase is restricted to  $\Phi_0$  or  $\Phi_2$  (never  $\Phi_1$  or  $\Phi_3$ ), governed by the Legendre symbol  $\left(\frac{2}{p}\right)$ , hence by  $p \pmod{8}$ .

## 1 Introduction

In [1], we proved that for  $p = 2n - 1$  prime, the interior  $\mathcal{I}_n^\circ = \{(n-1)^2 + 1, \dots, n^2 - 1\}$  is a punctured complete residue system modulo  $p$ , covering every class except  $r \equiv n^2 \pmod{p}$ . Since  $r$  is a quadratic residue,  $N^- - N^+ = 1$ .

At the end of [1], we asked whether analogous asymmetries hold for higher-order residue symbols. In this note, we answer affirmatively and go further: we identify the missing residue as  $r \equiv 4^{-1} \pmod{p}$  and use this to determine the deficient phase *exactly* for cubic and quartic residues via classical reciprocity laws.

## 2 The General Theorem

**Definition 2.1** ( $k$ -th power residue classes). Let  $p$  be an odd prime and  $k \geq 2$  with  $k \mid (p - 1)$ . Fix a primitive  $k$ -th root of unity  $\zeta \pmod{p}$ . The  $k$ -th power residue symbol partitions  $(\mathbb{Z}/p\mathbb{Z})^*$  into  $k$  classes  $\Phi_j = \{a : a^{(p-1)/k} \equiv \zeta^j \pmod{p}\}$ , each of cardinality  $(p - 1)/k$ .

**Theorem 2.2** (Higher-order phase deficit). *Let  $n \geq 2$  with  $p = 2n - 1$  prime, and let  $k \geq 2$  with  $k \mid (p - 1)$  and  $(p - 1)/k \geq 2$ . Then the interior  $\mathcal{I}_n^\circ$  contains:*

- (i) *exactly  $(p - 1)/k - 1$  elements in the class  $\Phi_j$  containing  $r \equiv 4^{-1} \pmod{p}$  (the deficient phase),*
- (ii) *exactly  $(p - 1)/k$  elements in each  $\Phi_i$  with  $i \neq j$ ,*
- (iii) *exactly 1 element divisible by  $p$ .*

*Proof.* The interval bounds satisfy  $(n-1)^2 = n^2 - (2n-1) = n^2 - p$ , so  $(n-1)^2 \equiv n^2 \pmod{p}$ . Thus  $\mathcal{I}_n^\circ$  consists of exactly  $p-1$  consecutive integers, forming a complete residue system modulo  $p$  punctured at  $r \equiv n^2 \pmod{p}$ .

Since  $p = 2n-1$  gives  $2n \equiv 1$ , we have  $n \equiv 2^{-1}$  and  $r \equiv n^2 \equiv 4^{-1} \pmod{p}$ .

Among the  $p-1$  nonzero classes, each phase  $\Phi_j$  has  $(p-1)/k$  elements. Removing the single class  $r \in \Phi_j$  reduces  $|\Phi_j \cap \mathcal{I}_n^\circ|$  by 1.  $\square$

**Corollary 2.3** (Character sum identity). *For any nontrivial multiplicative character  $\chi$  of order dividing  $k$  modulo  $p$ :*

$$\sum_{x \in \mathcal{I}_n^\circ} \chi(x) = -\chi(r) = -\chi(4^{-1}) = -\overline{\chi(2)}^2. \quad (1)$$

*This is an exact evaluation.*

*Proof.*  $\sum_{x \in \mathcal{I}_n^\circ} \chi(x) = \sum_{\substack{a=1 \\ a \neq r}}^{p-1} \chi(a) = -\chi(r)$ , since  $\sum_{a=1}^{p-1} \chi(a) = 0$  for nontrivial  $\chi$ . Then  $\chi(4^{-1}) = \overline{\chi(4)} = \overline{\chi(2)^2} = \overline{\chi(2)}^2$ .  $\square$

### 3 Identifying the Deficient Phase

#### 3.1 $k = 2$ : Quadratic (Paper IX recovery)

Since  $4^{-1} = (2^{-1})^2$  is a perfect square,  $r \in \Phi_0$  always. This recovers  $N^- - N^+ = 1$ .

#### 3.2 $k = 3$ : Cubic residues ( $p \equiv 1 \pmod{3}$ )

**Proposition 3.1** (Cubic deficient phase). *The deficient cubic phase is determined by the cubic character of 2:*

$$\chi_3(r) = \chi_3(4^{-1}) = \overline{\chi_3(2)}^2 = \chi_3(2)^4 = \chi_3(2),$$

where the last step uses  $\chi_3^3 = 1$ .

Concretely: if 2 is a cubic residue modulo  $p$  ( $2^{(p-1)/3} \equiv 1$ ), then the deficit is in  $\Phi_0$ . Otherwise, the deficit is in  $\Phi_1$  or  $\Phi_2$  according to  $2^{(p-1)/3} \pmod{p}$ .

By classical algebraic number theory, 2 is a cubic residue modulo a prime  $p \equiv 1 \pmod{3}$  if and only if  $p$  is representable as  $p = x^2 + 27y^2$  for some integers  $x, y$ .

#### 3.3 $k = 4$ : Quartic residues ( $p \equiv 1 \pmod{4}$ )

**Proposition 3.2** (Quartic phase restriction). *The deficient quartic phase is always  $\Phi_0$  or  $\Phi_2$  (never  $\Phi_1$  or  $\Phi_3$ ). Specifically:*

$$\chi_4(r) = \chi_4(4^{-1}) = \chi_4(2)^{-2} = \chi_2(2)^{-1} = \left(\frac{2}{p}\right),$$

since  $\chi_4^2 = \chi_2$ . Therefore:

- $p \equiv 1 \pmod{8}$ :  $\left(\frac{2}{p}\right) = 1$ , deficit in  $\Phi_0$ .
- $p \equiv 5 \pmod{8}$ :  $\left(\frac{2}{p}\right) = -1$ , deficit in  $\Phi_2$ .

*Proof.*  $\chi_4(4^{-1}) = \chi_4(2^{-2}) = \chi_4(2)^{-2}$ . Since  $\chi_4^2 = \chi_2$ ,  $\chi_4(2)^{-2} = \chi_2(2)^{-1} = \left(\frac{2}{p}\right)$ , which equals  $\pm 1$ . Since  $\Phi_0$  corresponds to character value 1 and  $\Phi_2$  to  $-1$ , the deficit is confined to these two.  $\square$

## 4 Computational Illustration

Since Theorem 2.2 is an unconditional algebraic identity, its computational verification is a tautology. We display sample data to illustrate the mechanism.

Table 1: Phase deficit verification for  $k = 2, 3, 4, 5$ , all qualifying primes up to  $n = 1000$ .

| $k$           | Qualifying primes | Deficit confirmed |
|---------------|-------------------|-------------------|
| 2 (quadratic) | 549               | ✓ (all)           |
| 3 (cubic)     | 148               | ✓ (all)           |
| 4 (quartic)   | 146               | ✓ (all)           |
| 5 (quintic)   | 73                | ✓ (all)           |

Table 2: Cubic phase deficit ( $k = 3$ ). \* marks the deficient class. The final column shows  $2^{(p-1)/3} \bmod p$ .

| $n$ | $p$ | $ \Phi_0 $ | $ \Phi_1 $ | $ \Phi_2 $ | $2^{(p-1)/3}$ |
|-----|-----|------------|------------|------------|---------------|
| 7   | 13  | 4          | 3*         | 4          | $\omega$      |
| 10  | 19  | 6          | 5*         | 6          | $\omega$      |
| 16  | 31  | 9*         | 10         | 10         | 1             |
| 22  | 43  | 13*        | 14         | 14         | 1             |
| 34  | 67  | 22         | 22         | 21*        | $\omega^2$    |

Table 3: Quartic phase restriction ( $k = 4$ ). The deficit is always in  $\Phi_0$  or  $\Phi_2$ , governed by  $p \bmod 8$ .

| $n$ | $p$ | $p \bmod 8$ | $ \Phi_0 $ | $ \Phi_1 $ | $ \Phi_2 $ | $ \Phi_3 $ | Deficit  |
|-----|-----|-------------|------------|------------|------------|------------|----------|
| 9   | 17  | 1           | 3*         | 4          | 4          | 4          | $\Phi_0$ |
| 15  | 29  | 5           | 7          | 7          | 6*         | 7          | $\Phi_2$ |
| 21  | 41  | 1           | 9*         | 10         | 10         | 10         | $\Phi_0$ |
| 27  | 53  | 5           | 13         | 13         | 12*        | 13         | $\Phi_2$ |

## 5 Discussion

### 5.1 The role of $4^{-1}$

The identification  $r \equiv 4^{-1} \pmod{p}$  is the key algebraic insight. It reduces the study of the deficient phase to the classical problem of determining the  $k$ -th power residue character of 2 modulo  $p$ . For  $k = 2$ : trivially  $\Phi_0$  (since  $4^{-1}$  is a square). For  $k = 3$ : connects to cubic reciprocity and the representation  $p = x^2 + 27y^2$ . For  $k = 4$ : reduces to the Legendre symbol  $\left(\frac{2}{p}\right)$  via the supplementary law.

### 5.2 Simultaneous deficits

When  $p - 1$  is highly composite, the interval simultaneously exhibits compatible deficits across all dividing  $k$ , all from the single puncture at  $4^{-1}$ .

### 5.3 Open questions

**Open Question 5.1** (Cubic Oppermann analogue). For  $k = 3$ , does the right half of  $\mathcal{I}_n$  exhibit zero cubic skewness under some congruence condition on  $n$ ?

**Open Question 5.2** (Beyond Burgess for Legendre intervals). The identity (1) evaluates the full character sum exactly. For half-intervals, Burgess [3] gives  $O(p^{3/16+\varepsilon})$  bounds for short sums. Can the arithmetic structure of Legendre half-intervals yield improvements for these specific ranges?

### Data and code availability

Verification scripts are available at <https://github.com/Ruqing1963/higher-order-residue-deficits>.

### References

- [1] R. Chen, “Algebraic rigidity and quadratic residue asymmetry in Legendre intervals,” Zenodo, 2026. <https://zenodo.org/records/18706876>
- [2] R. Chen, “Oppermann’s parity law: quadratic residue symmetry breaking in half-intervals via the negation involution,” Zenodo, 2026. <https://zenodo.org/records/18707265>
- [3] D. A. Burgess, “On character sums and  $L$ -series. II,” *Proc. London Math. Soc.* (3) **13** (1963), 524–536.